



RossClean CHARGER

Liquid additive to caustic for CIP

RossClean CHARGER is a low foaming alkaline concentrated detergent additive to caustic for single step cleaning of Evaporators, Pasteurizers, etc.

CHARGER is a highly effective on removal of milk soils and deposits in even in hard water conditions. It also effective for cleaning of bienzymatic applications when used with caustic. Charger can be used at different concentrations but with different caustic levels, where soiling types vary for the most cost-effective cleaning results throughout the Dairy.

It can be used in a wide range of applications across the Food & Beverage sectors, e.g. Brewhouse CIP, Calcium Oxalate and Mold removal.

Key Benefits

- Low foaming under conditions of high pressure and turbulence
- Cost effective as it's a blend of highly concentrated liquids
- Effective on Organic & Inorganic deposits removal
- Single stage cleaning
- Wide range of cleaning application

Application Method

Use at the concentrations of 0.2-2% w/w (0.2-1.5% v/v) with diluted caustic (not caustic liquor directly) depending on the application, soiling and water hardness.

HTST the recommended caustic levels are 0.3-1% w/w and for Evaporators (whole milk) the caustic levels are 1-2% w/w and should be rinse thoroughly after use with portable water from contact surfaces.

Technical Data

Appearance	Clear pale yellow
pH (10% solution)	12.00 $\pm$ 1
Specific Gravity	1.298 $\pm$ 0.1
Solid Content	50.00% $\pm$ 2



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Product Compatibility

Compatible with materials commonly found in the food and beverage industry. In case of uncertainty specific tests should be undertaken.

Storage & Handling

Store in original closed containers or (where applicable) in an approved bulk tank, away from extreme temperatures. Full guidance on the handling and disposal of this product is provided in a separate Material Safety Data Sheet.

Declaration: Technical figures given in this data sheet are typical values based on internal laboratory measurements, following appropriate International Standard Methods, and should not be taken as a warranty or representation. It is the end user's responsibility to make sure that the product matches its requirements.

For more information on this product, please contact your local Buzil Rossari representative.

Buzil Rossari Private Limited

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Liquid additive to caustic for CIP

Food Grade Quality Certificate

Application in food & beverage processing facilities

Here we confirm that, **RossClean CHARGER** if used according to the instructions mentioned in the product information sheet or intended use, is suitable for cleaning in Beverage, Brewing, Dairy & Food Processing facilities and not to adulterate the food product in any manner during its use.

Essential condition is a thorough rinsing with clear portable water of surfaces intended to get in direct contact with foodstuff.

The remaining residuals that may contaminate or adulterate food contacting cleaned surfaces are expected to be negligible, following a potable water rinse. Therefore, the potential ingestion of residues resulting from the approved use of these types of products does not represent a significant health hazard after our assessment of the product as suitable for use under the intended conditions.



Thanking you,

Yours faithfully,

For Rossari Biotech Ltd.

Please note that this is not a regulatory assessment and should not be construed as an endorsement of regulatory compliance.

MATERIAL SAFETY DATA SHEET

according to REGULATION (EU) No 453/2010, annex I

RossClean Charger

Date of issue: 10.11.2018

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1. Identification of the mixture and of the company / undertaking

1.1. Product identifier

RossClean Charger

1.2. Relevant identified uses of the mixture and uses advised against

Professional cleaning/maintenance product for food & beverage industries.

1.3. Details of the supplier of the safety data sheet

Manufacturer:

Manufactured under licence from buzil
ROSSARI BIOTECH LIMITED
Plot No. 10 & 11, Survey No.90/1/10 & 90/1/11/1,
Kumbharwadi, Village Naroli
Silvassa-396235
Dadra and Nagar Haveli (UT)
India

e-mail info@rossarimail.com

www.buzil-rossari.com & www.rossari.com

1.4. Emergency telephone number

+91-22-61233800/61233887

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2. Hazards identification

2.1. Classification of the mixture

H302: Harmful if swallowed

H319: Eye irritation, category 2

2.2. Label elements (acc. to 1999/45/EC)



H302: Harmful if swallowed

H319: Eye irritation, category 2

R 34 - Causes burns.

R 36 - Irritating to eyes.

R 37 - Irritating to respiratory system.

Classification system : The classification complies with current EC regulations on dangerous substances and preparations.

2.3. Other hazards

The product does not contain substances which are classified as persistent, bioaccumulative and toxic (PBT) or very persistent and very bioaccumulative (vPvB) according to 1907/2006/EG annex XIII.

Foam formation after vomiting bears risk of suffocation.

3. Composition / information on ingredients

Chemical characterization

Description : Mixture in water of non-dangerous ingredients and substances listed below.

Hazardous ingredients :

CAS: 64-02-08 Tetrasodium ethylenediamine tera-acetate Xn, Xi; R22-36 ≥30%

EINECS: 200-573-9

CAS: 1312-76-1 Dipotassium metasilicate C; R34-37 <5%

EINECS: 215-199-1

Ingredients according to EC 89/542/EEC :

EDTA - Sodium salt ≥30%

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Anionic surfactants, phosphonates, phosphates, non-ionic surfactants

<5%

4. First aid measures

General information : Symptoms may occur after several hours, therefore medical observation for atleast 48hrs after the incident is required

Inhalation : Not applicable

Skin contact : Wash skin immediately and thoroughly with water. Remove contaminated clothing & wash before re -use. Obtain medical attention if symptoms develop.

Eye contact : Wash immediately with copious amounts of water and obtain medical attention.

Ingestion : Rinse out mouth and drink one or two glasses of water(or milk). Obtain medical attention.

5. Firefighting measures

Suitable extinguishing media: CO₂, extinguishing powder or water jet. Fight larger fires with water jet or alcohol resistant foam.

Protective equipment: No special measures required

6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

No special equipments required

6.2. Environmental precautions:

Dilute with plenty of water

6.3. Methods and materials for containment and cleaning up

Collect mechanically. Recover for use if possible. Or transfer to suitable non-metallic container for disposal at the approved site. Dispose of the collected material according to the regulations. Ensure adequate ventilation.

6.4. Reference to other sections

Follow advices and protective measures mentioned in section 7 and 8.

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7. Handling and storage

7.1. Precautions for safe handling

(See sections 8 and 15) Use common rules for working with chemicals. Prevent formation dust. Add product to water and not vice-versa while preparing solutions

Information about protection against explosions and fires: No special measures required.

7.2. Conditions for safe storage, including any incompatibilities

Store in original closed container under dry conditions. Avoid extreme temperature.
Store in accordance to local legislations

7.3. Specific end uses

None

8. Exposure controls / personal protection

Additional information about design of technical systems : No further data; see section 7.

Additional information : The lists that were valid during the compilation were used as basis.

Personal Protection equipment :

General protective and hygienic measures :

Keep away from foodstuffs, beverages and food. Take off immediately all contaminated clothing.

Wash hands during breaks and at the end of work. Avoid contact with the eyes and skin

Breathing equipment: Personal protection is not normally required

Protection of hands: Protective gloves recommended..

Eye protection: Safty glasses recommended during handling of the product

9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance: Clear to pale yellow liquid

Odour: Characteristic

pH-value(10%): 12.00 ± 1

Melting point / melting range: ND

Boiling point / boiling range: ND

Density (25 °C): ND

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Solubility in / miscibility with water: Miscible in water

Viscosity (25 °C): ND

Specific gravity: 1.298 ± 0.1

10. Stability and reactivity

10.1. Reactivity

No extraordinary reactivity known.

10.2. Chemical stability

The product is stable under normal conditions (room temperature).

10.3. Possibility of hazardous reactions

Reacts with acids.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

No hazardous incompatibilities known.

10.6. Hazardous decomposition products

Toxic fumes of Na_xO_x

11. Toxicological information

Information on toxicological effects

Acute toxicity:

LD50: 200 - 2000mg/kg

Primary irritant effect:

Skin contact: Causes irritation

Eye contact : Causes irritation

Ingestion : Causes irritation .

Inhalation : Severe irritant, may cause pulmonary oedema.

Sensitisation : No sensitizing effect known.

Irritation, corrosion, sensitisation

Classification of the product according to 1999/45/EG: see section 2.

Additional toxicological information: EC classification: Irritant

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12. Ecological information

General Notes: Must not reach sewage water or drainage ditch undiluted or unneutralized. When used for its intended purpose, this product should not cause adverse effects in the environment

13. Disposal considerations

Product Recommendation: Must not be disposed of together with domestic waste. Do not allow undiluted product to reach sewage system.

Uncleaned Packagings Recommendation : Dispose of observing official regulations.

Recommended cleaning agent : Water, if necessary with cleaning agent.

14. Transport information

Land transport ADR/RID (cross-border)

ADR/RID-GGVSE Class : 8

Kemler No. 80

UN Number : 3267

Item Number/ sub-group : III

Label : 8

Packaging Group : III

Proper Shipping Name : CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.

Maritime transport IMDG:

IMDG class: 8

UN Number : 3267

Item Number/ sub-group : III

Label : 8

Packaging Group : III

Proper Shipping Name : CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.

Air transport ICAO and IATA-DGR

IMDG class: 8

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UN Number : 3267

Item Number/ sub-group : III

Label : 8

Packaging Group : III

Proper Shipping Name : CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.

15. Regulatory information

Classification according to EC guidelines : The product has been classified and labeled in accordance with current EC Directives.

Symbol and hazard designation of product:



16. Other information

The information in this document is based on our best present knowledge. However, it does not constitute a guarantee or any specific product features and does not establish a legally binding contract.



Activity Method

Product	RossClean CHARGER	Issued On	01/03/2018
Application	Caustic detergent for single stage CIP	Next Review On	01/03/2023

Principle

Alkalinity in the sample is neutralized by the standard acid and the corresponding acidity consumed is used to calculate the activity of the sample.

Reagents & Apparatus

Phenolphthalein Indicator, 1N Hydrochloric acid, DM Water
Conical Flask, Burette, Measuring Cylinder

Method

- Weight 2-3gms of the product in the flask with 100mL of DM Water
- Add 1-2 drops of Phenolphthalein Indicator and mix it well
- Titrate the solution with 1N Hydrochloric acid till the solution colour changes from Pink to colourless
- Pink color should not appear for a minimum 15 seconds
- Take the burette reading (BR) and do the calculation as shown below,

Calculation

Activity % of CHARGER = $(BR \times \text{Normality of HCL} \times 49) / \text{Weight of the sample}$

% Activity should be 95% or above



User Concentration Method

Method - 1

Principle

The chelating agent in the sample is estimated by titrating with the standard Metal Ion solution at alkaline pH, which corresponds to the concentration of the sample.

Reagents & Apparatus

SU reagent, Eriochrome Black T Indicator, Ammonia Buffer, DM Water
Conical Flask, Burette, Measuring Cylinder

Method

- Take 10mL of the user solution in the flask with 50mL of DM Water
- Add 1-3 mL of Ammonia Buffer in the solution till pH reaches to 11
- Titrate against the SU reagent till the blue shade changes to wine red color
- Take note of burette reading (BR) and calculate the user concentration as per below mentioned calculation

Calculation

Concentration % of Charger = $(BR \times 0.625)$

Method - 2

Principle

Performance of RossClean CHARGER through total EDTA

Reagents

0.1 N Copper Sulfate, PAN indicator (0.1% in ethanol), Phenolphthalein Indicator, Acetate buffer solution (pH 4.66),
1 N Hydrochloric acid

Method

pH Adjustment

Up to 2-3 drops of the phenolphthalein indicator to 50 ml of use. Titrate with hydrochloric acid until the sample becomes colorless. Add 20 ml of acetate buffer.

Determination

Up to 1 ml of PAN Indicator reading to the sample (the base is changed to yellow) and titrate to a purple / red end point with copper sulfate.

Calculation

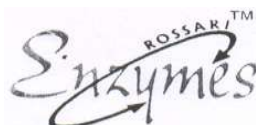
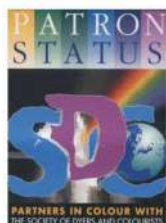
Concentration of Charger % (w / w) = $\text{titer (ml)} \times 0.23$

Concentration of Charger % (v / v) = $\text{titer (ml)} \times 0.17$

TEST REPORT

Product revision No. & Date		:		
Product		:	ROSSCLEAN CHARGER	
Batch No		:	BR 181367	
Manufacturing Date		:	07.03.2019	
S.No.	PARAMETER		STANDARD	BATCH
1	Physical Form	:	Light yellowish clear liquid	Light yellowish clear liquid
2	pH 10% Solution	:	12.00 +/-1	12.20
3	Miscibility	:	Miscible in water	Miscible in water
4	Specific Gravity	:	1.298 +/- 0.1	1.273
5	Solid content (110°C, 2hrs)	:	50.00% +/- 2	51.61%
6	Odour	:	Characteristic	Characteristic

Note :- This is computer generated print no signature is required.



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